

Lab 1: Fragrance Soap Making

Introduction

Behind every beauty product on your shelf is a story of chemistry. Soap, one of the oldest products in any skincare routine, is no exception. In this experiment, we explore how a simple reaction between oil and sodium hydroxide creates soap. Understanding the science behind what we put on our bodies is the first step to becoming an Elementalist.

Soap is produced through a chemical reaction known as **saponification**, in which triglycerides (fats or oils) react with a strong base such as sodium hydroxide (NaOH) to form glycerol and fatty acid salts (soap). In this reaction, the ester bonds in the triglyceride molecules are hydrolyzed by hydroxide ions, breaking the molecule into glycerol and long-chain carboxylate ions. These carboxylate ions associate with sodium ions to form soap. The overall reaction can be represented as:



Mechanism

The hydroxide ion acts as a nucleophile and attacks the carbonyl carbon (C=O) of each ester group. The carbonyl double bond opens to form a short-lived tetrahedral intermediate, which then collapses: the double bond re-forms and the alkoxide leaving group is expelled. A final proton transfer leaves a carboxylate ion and an alcohol. Each triglyceride carries three ester groups, so one molecule of fat consumes three hydroxide ions and releases three soap molecules and one molecule of glycerol.

Why Each Oil Matters

Every oil in the recipe brings a different blend of fatty acids, and those fatty acids decide how the finished bar behaves.

- **Coconut oil** is rich in lauric acid, which gives a hard bar and a big, fluffy lather.
- **Olive oil** is mostly oleic acid: a gentle, conditioning bar that lathers slowly.
- **Palm oil** adds palmitic acid for hardness and a long-lasting bar.
- **Castor oil** contributes ricinoleic acid, which stabilises the lather into a creamy foam.
- **Shea butter** (optional) leaves some of its unsaponifiable fraction in the bar, which is why it feels moisturising.

Superfat

Recipes are calculated with slightly less sodium hydroxide than the oils could consume, usually 5 to 8 percent less. This **superfat** guarantees that no free lye remains in the bar and leaves a small amount of unreacted oil behind to soften the skin. Our recipe uses 85g of NaOH for 610g of oils, which lands inside that range.

Working Safely With Lye

- Wear goggles and gloves from the moment the NaOH container is opened until everything is washed up.
- Always add the lye to the water, never the water to the lye. Adding water to solid NaOH can boil and spatter.
- Dissolving NaOH is strongly exothermic; the solution will reach about 80°C and give off fumes. Mix under ventilation.
- Keep a bottle of plain water nearby. Rinse any skin contact for several minutes; vinegar is not a substitute for rinsing.

Reaching Trace

Saponification begins as soon as the lye solution meets the oils, but at first the two phases barely mix. Blending forces the oil into tiny droplets, increasing the surface where the reaction can happen. As soap forms, it acts as an emulsifier and the batter thickens. **Trace** is the point where a drizzle of batter rests on the surface for a moment before sinking back in. Pour at light trace for swirls and at medium trace for a smooth, even bar.

Adding Fragrance

Fragrance and essential oils are added at trace so they are not exposed to the hottest part of the reaction. Some fragrances accelerate trace or cause the batter to seize, so stir them in by hand with a spatula rather than with the blender.

Curing

The bars can be unmolded after 24 to 48 hours, but saponification continues for several days and the bars keep losing water for weeks. Cure them on a rack with air on every side for four to six weeks. A cured bar is harder, milder and lasts longer in the shower.

Review Questions

- Why does one triglyceride molecule produce three soap molecules?
- What would happen to the finished bar if the recipe had no superfat?
- Why is the lye always added to the water and not the reverse?
- Which oil in the recipe contributes most to a hard bar with a big lather?